

Project Design Phase
Proposed Solution Template

Date	1 July 2026
Team ID	
Project Name	
Maximum Marks	2 Marks

Proposed Solution Template – HouseRent (MERN Stack House Rental Platform)

Project Information

S.No.	Parameter	Description
1	Project Title	HouseRent – MERN Stack House Rental Platform
2	Domain	Web Application / Real Estate
3	Technology Stack	MongoDB, Express.js, React.js, Node.js, JWT, Cloudinary, Nodemailer
4	Target Users	House Seekers (Tenants), Property Owners, Administrators
5	Objective	To provide a secure, centralized, and user-friendly platform for renting houses and managing property listings.

1. Problem Statement (Problem to be Solved)

Problem Statement (Problem to be solved)

House seekers often face difficulty finding genuine rental properties due to fake or outdated listings. Property owners struggle to manage listings and bookings efficiently, while administrators lack a centralized system to verify owners and monitor platform activities.

2. Idea / Solution Description

Idea / Solution Description

Develop **HouseRent**, a MERN Stack-based rental platform that allows users to search and book verified rental properties, enables owners to manage property listings and bookings, and provides administrators with tools to approve owners, manage users, properties, and bookings. The platform includes JWT authentication, Cloudinary image uploads, and email-based password recovery for security and convenience.

3. Novelty / Uniqueness

Novelty / Uniqueness

- Admin approval for property owners before listing properties.
 - Secure JWT-based authentication with role-based access control.
 - Cloudinary support for multiple property image uploads.
 - Email-based password reset using Nodemailer.
 - Separate dashboards for Users, Owners, and Admin with dedicated functionalities.
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4. Social Impact / Customer Satisfaction

Social Impact / Customer Satisfaction

- Provides verified and trustworthy rental property listings.
 - Reduces fraudulent property advertisements.
 - Saves time for tenants searching for rental houses.
 - Helps owners manage properties efficiently.
 - Improves customer confidence through secure authentication and admin verification.
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5. Business Model (Revenue Model)

Business Model (Revenue Model)

- Premium subscription plans for property owners to feature listings.
 - Featured property advertisements.
 - Commission on successful property bookings (future enhancement).
 - Partnerships with real estate agencies and property management companies.
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6. Scalability of the Solution

Scalability of the Solution

- Can support thousands of users and property listings using the MERN Stack.
- Easy integration of future features such as online payments, reviews, ratings, live chat, notifications, AI-based property recommendations, and analytics dashboard.
- Cloud-based image storage and MongoDB ensure efficient scalability and performance.